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Studierichting: Informatica
Oefeningenreeks: 4A nr 24.3

Bereken $\int \frac{x-1}{x^2} \ln x dx$

Splits de breuk:

$$\begin{aligned}\int \frac{x-1}{x^2} \ln x dx &= \int \left(\frac{1}{x} - \frac{1}{x^2} \right) \ln x dx \\ &= \int \frac{\ln x}{x} dx - \int \frac{\ln x}{x^2} dx\end{aligned}$$

Eerste integraal: substitueer $t = \ln x$, dan is $dt = \frac{dx}{x}$

$$\begin{aligned}\int \frac{\ln x}{x} dx &= \int t dt \\ &= \frac{t^2}{2} + c \\ &= \frac{\ln^2 x}{2} + c\end{aligned}$$

Tweede integraal: partiële integratie

Stel

$$\begin{cases} u = \ln x \\ dv = \frac{dx}{x^2} \end{cases} \Rightarrow \begin{cases} du = \frac{dx}{x} \\ v = -\frac{1}{x} \end{cases}$$

dan wordt

$$\begin{aligned}\int \frac{\ln x}{x^2} dx &= -\frac{\ln x}{x} - \int \left(-\frac{1}{x} \right) \cdot \frac{dx}{x} \\ &= -\frac{\ln x}{x} + \int \frac{dx}{x^2} \\ &= -\frac{\ln x}{x} - \frac{1}{x} + c\end{aligned}$$

Samen:

$$\begin{aligned}
I &= \frac{\ln^2 x}{2} - \left(-\frac{\ln x}{x} - \frac{1}{x} \right) + c \\
&= \frac{\ln^2 x}{2} + \frac{\ln x}{x} + \frac{1}{x} + c
\end{aligned}$$

References